

KAUSHIK SENTHOOR

University of Sheffield, UK

 ksenthoor.github.io
 k.s.rajendran@sheffield.ac.uk
 

ABOUT ME

I am a postdoctoral researcher in Dr. Yingkai Ouyang's group at University of Sheffield. My current research work is on the topic of error correction codes for quantum metrology. I received my Ph.D. from the Indian Institute of Technology Madras. My doctoral research focused on how to reduce the communication cost during secret recovery in quantum secret sharing schemes.

RESEARCH INTERESTS

Classical and quantum error correction codes, information theory, quantum cryptography and signal processing

EDUCATION

Ph.D. in Electrical Engineering Jan 2016 - Jul 2023

Indian Institute of Technology Madras

Advisor : Prof. Pradeep Sarvepalli

Thesis : Communication Efficient Quantum Secret Sharing Using Quantum Codes

M.E. in Telecommunication Engineering Aug 2012 - Jul 2014

Indian Institute of Science Bangalore

Advisor : Prof. P Vijay Kumar

Thesis : Storage Overhead vs. Repair Bandwidth Tradeoff in Exact Repair Regenerating Codes

B.Tech. in Electronics & Communication Engineering Aug 2008 - Jul 2012

Amrita School of Engineering Coimbatore

Group project : Compressed sensing in ECG signals

WORK EXPERIENCE

Postdoctoral researcher Apr 2026 - Present

University of Sheffield

Postdoctoral researcher Nov 2023 - Nov 2025

QuTech, Delft University of Technology

Quantum network applications

Engineer Aug 2014 - May 2015

Mobile-Video team, Ittiam systems, Bangalore

Development of video codecs in H264 and HEVC standards

Project Associate Jun 2015 - Dec 2015

Codes and Signal Design lab, Indian Institute of Science Bangalore

Project for intrusion detection using Wireless Sensor Networks

PUBLICATIONS

Preprint

- [1] K. Senthooor, "[Entanglement Cost of Erasure Correction in Quantum MDS Codes](#)," *e-print quant-ph/2505.20284*, 2025.

Journal papers

- [2] K. Senthoo and P. K. Sarvepalli, “[Communication Efficient Quantum Secret Sharing via Extended CSS Codes](#),” *IEEE Journal on Selected Areas in Communications*, vol. 42, no. 7, pp. 1818–1829, 2024.
- [3] K. Senthoo and P. K. Sarvepalli, “[Theory of Communication Efficient Quantum Secret Sharing](#),” *IEEE Transactions on Information Theory*, vol. 68, no. 5, pp. 3164–3186, 2022.
- [4] K. Senthoo and P. K. Sarvepalli, “[Communication efficient quantum secret sharing](#),” *Physical Review A*, vol. 100, no. 5, p. 052313, 2019.
- [5] B. Sasidharan, N. Prakash, M. N. Krishnan, M. Vajha, K. Senthoo, and P. V. Kumar, “[Outer bounds on the storage-repair bandwidth trade-off of exact-repair regenerating codes](#),” *International Journal of Information and Coding Theory*, vol. 3, no. 4, pp. 255–298, 2016.

Conference proceedings

- [6] K. Senthoo and P. K. Sarvepalli, “[Concatenating Extended CSS Codes for Communication Efficient Quantum Secret Sharing](#),” in *Proc. 2023 International Symposium on Topics in Coding (ISTC), Brest, France*.
- [7] K. Senthoo, B. Sasidharan, and P. V. Kumar, “[Improved layered regenerating codes characterizing the exact-repair storage-repair bandwidth tradeoff for certain parameter sets](#),” in *Proc. 2015 IEEE Information Theory Workshop (ITW), Jerusalem, Israel*.
- [8] B. Sasidharan, K. Senthoo, and P. V. Kumar, “[An improved outer bound on the storage-repair-bandwidth tradeoff of exact-repair regenerating codes](#),” in *Proc. 2014 IEEE International Symposium on Information Theory (ISIT), Honolulu, Hawaii, USA*.

PRESENTATIONS

- Talk at AQIS 2018, Nagoya, Japan.
- Poster at Indo-German Symposium on Quantum Science and Technologies 2020, IIT Madras, India.
- Poster at Quantum Information Processing 2025, Raleigh, USA.

PROGRAMMING

I have good expertise in C and MATLAB. I also have exposure to programming in C++.

TEACHING EXPERIENCE

Applied Linear Algebra I, Jul - Nov 2018

- One of four teaching assistants for a class of about 50 students. Conducted weekly tutorial sessions and discussed solutions for problem sets. Evaluated video presentations by students on recent research articles.

Information Theory (online mode), Jul - Nov 2022

- One of seven teaching assistants for a class of about 60 students. Conducted weekly tutorial sessions for a group of 10 students. Partly evaluated the mini-quiz and final exam papers.

I worked as a teaching assistant also in courses such as Error Control Codes, Modern Coding Theory and Advanced Topics in Quantum Information during my Ph.D. programme.

ACADEMIC SERVICES

- Reviewed papers for journals such as IEEE Transactions on Information Theory, IEEE Journal on Selected Areas in Information Theory and ACM Transactions on Quantum Computing.
- Regularly review papers for national and international research conferences such as NCC (India), IEEE ISIT, IEEE ISITA, TQC and QCrypt.
- Partly organized the transportation of the speakers in JTG Summer School 2019 at IIT Madras.
- Helped with event photography in Electrical Engineering Symposium 2018 at IIT Madras.